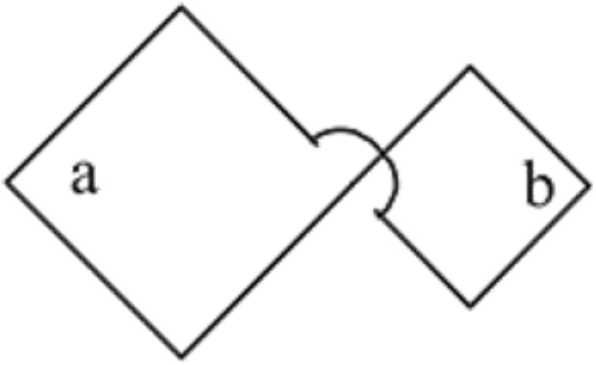
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A plane loop, shaped as two squares of sides a $= 1$ m and b $= 0.4$ m is introduced into a uniform magnetic field $\perp$ to the plane of loop. The magnetic field varies as $\mathrm{B} = 10^{-3}\sin(100\mathrm{t})\mathrm{T}$. The amplitude of the current induced in the loop if its resistance per unit length is $\mathrm{r} = 5\ \mathrm{m}\Omega\mathrm{m}^{-1}$ is



(A) 2 A

(B) 3 A

(C) 4 A

(D) 5 A

Q.2

A coil of resistance 400Ω is placed in a magnetic field. If the magnetic flux $\phi(wb)$ linked with the coil varies with time $t(\mathrm{sec})$ as $\phi = 50t^2 + 4$. The current in the coil at $t = 2\mathrm{sec}$ is :

(A) 0.5 A

(B) 0.1 A

(C) 2 A

(D) 1 A

Q.3

A coil having n turns and resistance $R\Omega$ is connected with a galvanometer of resistance $4R\Omega$. This combination is moved in time second from a magnetic flux ϕ_1 weber to ϕ_2 weber. The induced current in the circuit is

(A) $-\frac{\phi_2-\phi_1}{5Rnt}$

(B) $-\frac{n(\phi_2-\phi_1)}{5Rt}$

(C) $-\frac{(\phi_2-\phi_1)}{Rnt}$

(D) $-\frac{n(\phi_2-\phi_1)}{Rt}$

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Q.4

A conducting disc of conductivity σ has a radius a and thickness 't'. If the magnetic field B is applied in a direction perpendicular to the plane of the disc changes with time at the rate of $\frac{dB}{dt} = \alpha$. Calculate the power dissipated in the disc due to the induced current.

(A) $\frac{\pi t \sigma a^4}{8}\alpha^2$

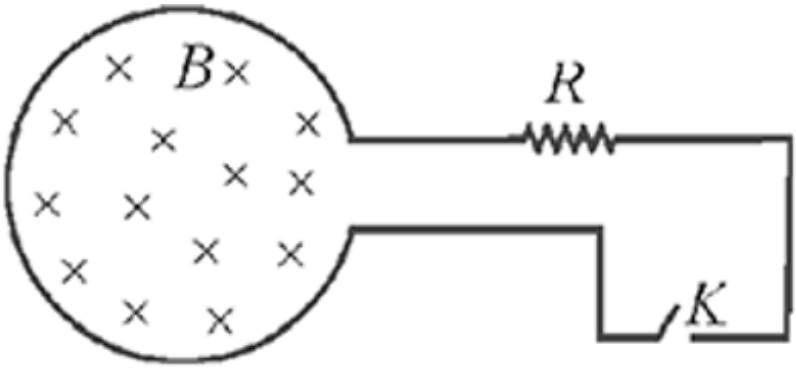
(B) $\frac{\pi t \sigma a^4}{4}\alpha^2$

(C) $\frac{\pi t \sigma a^4}{2}\alpha^2$

(D) $\frac{2\pi t \sigma a^4}{3}\alpha^2$

Q.5

Shown in the figure is a circular loop of radius r and resistance R. A variable magnetic field of induction $B = B_0 e^{-t}$ is established inside the coil. If the key (K) is closed, the electrical power developed right after closing the switch is equal to



(A) $\frac{B_0^2 \pi r^2}{R}$

(B) $\frac{B_0 10 r^3}{R}$

(C) $\frac{B_0^2 \pi^2 r^4 R}{5}$

(D) $\frac{B_0^2 \pi^2 r^4}{R}$

Q.6

A current carrying infinitely long wire is kept along the diameter of a circular wire loop, without touching it, the correct statement(s) is(are)

I. The emf induced in the loop is zero if the current is constant.

II. The emf induced in the loop is finite if the current is constant.

III. The emf induced in the loop is zero if the current decreases at a steady rate.

(A) I only

(B) II only

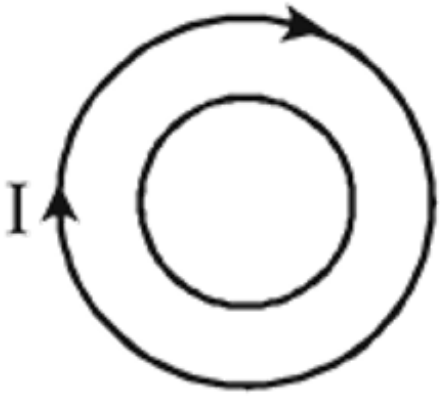
(C) I and II

(D) I, II and III

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Q.7

Two different wire loops are concentric and lie in the same plane. The current in the outer loop (I) is clockwise and increases with time. The induced current in the inner loop



(A) is clockwise

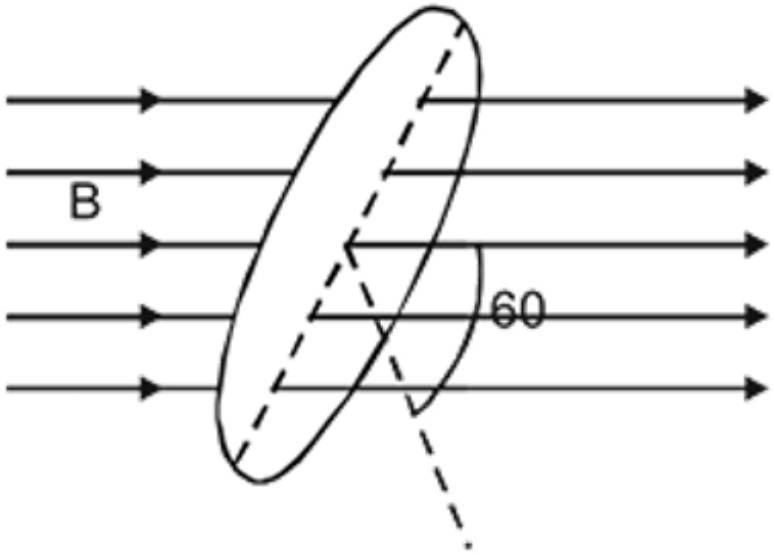
(B) is zero

(C) is counter clockwise

(D) has a direction that depends on the ratio of the loop radii

Q.8

Fig shown below represents an area $\mathrm{A} = 0.5\ \mathrm{m}^2$ situated in a uniform magnetic field $\mathrm{B} = 2.0$ weber/ m^2 and making an angle of 60° with respect to magnetic field.



The value of the magnetic flux through the area would be equal to

(A) 2.0 weber

(B) $\sqrt{3}$ weber

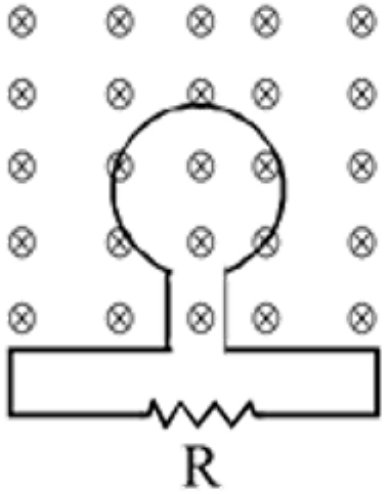
(C) $\sqrt{3}/2$ weber

(D) 0.5 weber

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Q.9

In the figure the flux through the loop perpendicular to the plane of the coil and directed into the paper is varying according to the relation $\phi = 6t^2 + 7t + 1$ where ϕ is in milliweber and t is in second. The magnitude of the emf induced in the loop at $t = 2$ s and the direction of induced current through R are θ



(A) 39mV; right to left

(B) 39mV; left to right

(C) 31mV; right to left

(D) 31mV; left to right

Q.10

Magnetic flux linked with a stationary loop of resistance R varies with respect to time during the time period T as follows:

$\phi = at(T - t)$

The amount of heat generated in the loop during that time (inductance of the coil is negligible) is

(A) $\frac{aT}{3R}$

(B) $\frac{a^2 T^2}{3R}$

(C) $\frac{a^2 T^2}{R}$

(D) $\frac{a^2T^3}{3R}$

Q.11

A conducting circular loop is placed in a uniform magnetic field of 0.04 T with its plane perpendicular to the magnetic field. The radius of the loop starts shrinking at 2 mm/s. The induced emf in the loop when the radius is 2 cm is

(A) $4.8\pi\mu\text{V}$

(B) $0.8\pi\mu\text{V}$

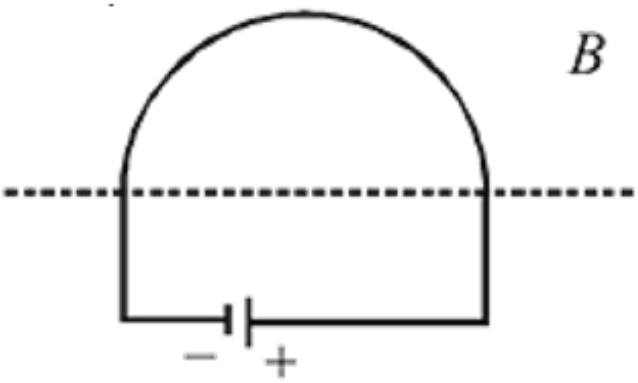
(C) $3.2\pi\mu\text{V}$

(D) $1.6\pi\mu\text{V}$

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Q.12

The figure shows a conducting loop consisting of half circle of area $A = 0.06\ \text{m}^2$ and three straight segments. The half circle lies in a uniform changing magnetic field $B = 4t^2 + 2t + 5$ (SI unit), where t is the time in second. An ideal battery $E = 2\ \text{V}$ is connected as shown and the total resistance of the wire is 2Ω. The net current in the loop is at $t = 5$ second is:



(A) $1\ \text{A}$

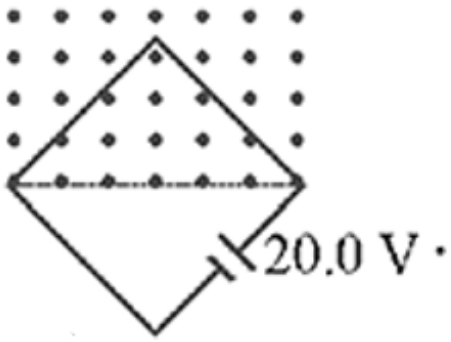
(B) $1.5\ \text{A}$

(C) $0.26\ \text{A}$

(D) $0.10\ \text{A}$

Q.13

A square loop with $2.0\ \text{m}$ sides is perpendicular to a uniform magnetic field, with half the area of the loop in the field is shown in figure. The loop contains a $20.0\ \text{V}$ battery with negligible internal resistance. If the magnitude of the field varies with time according to $B = 0.042 - 0.87t$, with B in tesla and t in second. The net emf of the circuit is:



(A) $20.0\ \text{V}$

(B) $18.26\ \text{V}$

(C) $21.74\ \text{V}$

(D) None of thes

Q.14

A coil of circular cross-section having 1000 turns and $4\ \text{cm}^2$ face area is placed with its axis parallel to a magnetic field which decreases by $10^{-2}\ \text{Wb m}^{-2}$ in $0.01\ \text{s}$. The e.m.f. induced in the coil is:

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(A) 400mV

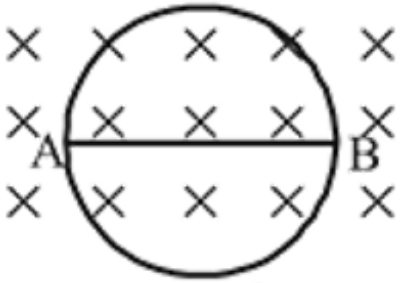
(B) 200mV

(C) 4mV

(D) 0.4mV

Q.15

The radius of the circular conducting loop shown in fig. is R. Magnetic field is decreasing at a constant rate α. Resistance per unit length of the loop is ρ.



Then, the current in wire AB is $(\text{AB}$ is one of the diameters)

(A) $\frac{\text{R}\alpha}{2\text{p}}$ from A to B

(B) $\frac{\text{R}\alpha}{2\text{p}}$ from B to A

(C) $\frac{\text{R}\alpha}{\text{o}}$ from A to B

(D) 0

Q.16

A flexible wire loop in the shape of a circle has radius that grows linearly with time. There is a magnetic field perpendicular to the plane of the loop that has a magnitude inversely proportional to the distance from the center of the loop, $\text{B}(\text{r}) \propto \frac{1}{\text{r}}$. How does the emf E vary with time?

(A) $\text{E} \propto \text{t}^2$

(B) $\text{E} \propto \text{t}$

(C) $\text{E} \propto \sqrt{\text{t}}$

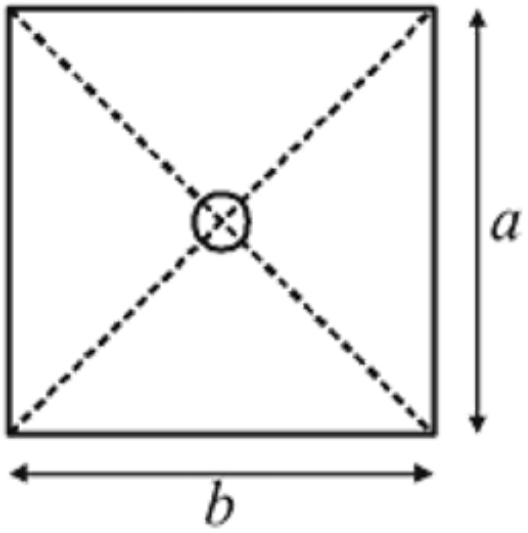
(D) E is constant

Q.17

A rectangular coil has a long straight wire passing through its centroid perpendicular to its plane as shown. If current through the wire varies as $i = i_0 \sin \omega t$, induced current in the coil will be (Given $R =$ Resistance of

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the coil)



(A) $\frac{i_0 \sin \omega t}{R}$

(B) $\frac{\pi a \sin \omega t}{bR}$

(C) zero

(D) $\frac{\pi a \cos \omega t}{bR}$

Q.18

Two identical cycle wheels (geometrically) have different number of spokes connected from center to rim. One is having 20 spokes and the other having only 10 (the rim and the spokes are resistanceless). One resistance of value R is connected between center and rim. The current in R will be

(A) double in the first wheel than in the second wheel

(B) four times in the first wheel than in the second wheel

(C) will be double in the second wheel than that of the first wheel

(D) will be equal in both these wheels

Q.19

The magnetic field in a region is given by $B = B_0 \left(1 + \frac{x}{a}\right) \hat{k}$.
A square loop of edge-length d is placed with its edges along the x and y -axes. The loop is moved with a constant velocity $v = v_0 \hat{i}$. The emf induced in the loop is :

(A) zero

(B) $\mathrm{v_0\ B_0\ d}$

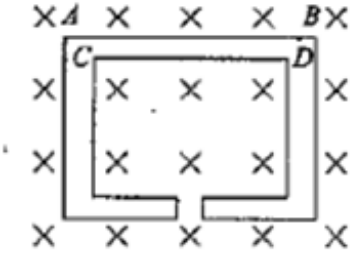
(C) $\frac{\mathrm{v_0\ B_0\ d^2}}{\mathrm{a}}$

(D) $\frac{\mathrm{v_0\ B_0\ d^3}}{\mathrm{a^2}}$

Q.20

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A wire is bent to form the double loop shown in figure. There is a uniform magnetic field directed into the plane of the loop. If the magnitude of this field is decreasing, the current will flow from



(A) a to b and c to d

(B) b to a and d to c

(C) a to b and d to c

(D) b to a and c to d

Answers & Solutions

Q.1 Answer:

3 A

Solution:

$\phi(\text{ flux linked }) = a^2 B\cos 0^\circ - b^2 B\cos 180^\circ$

$\mathrm{E} = -\frac{\mathrm{d}\phi}{\mathrm{dt}} = -\left(\mathrm{a}^2 - \mathrm{b}^2\right)\frac{\mathrm{dB}}{\mathrm{dt}}$

$= \left(a^2 - b^2\right) B_0\omega\cos\omega t$

where $\mathrm{B} = \mathrm{B}_0, \sin\omega\mathrm{t}, \mathrm{B}_0 = 10^{-3}\ \mathrm{T}, \omega = 100$

$\therefore \quad I_{\max} = \left(a^2 - b^2\right)\frac{B_0\omega}{R}$

and $\quad R = (4a + 4b)r = 4(a + b)r$

$I_{\max} = \frac{(\mathrm{a}-\mathrm{b})\mathrm{B}_0\omega}{4\mathrm{r}} = \frac{(1-0.4)\times 10^{-3}\times 100}{4\times 5\times 10^{-3}} = 3\ \mathrm{A}$

Q.2 Answer:

0.5 A

Solution:

The magnitude of induced emf is

$|\varepsilon| = \left|-\frac{\mathrm{d}\phi}{\mathrm{dt}}\right| = 100\mathrm{t}$

By Ohm's law, current at $\mathrm{t} = 2$ **is**

$\mathrm{I} = \frac{\varepsilon}{\mathrm{R}} = 100\mathrm{t}/400 = 100(2)/400 = 0.5\ \mathrm{A}$

Q.3 Answer:

$-\frac{n(\phi_2-\phi_1)}{5Rt}$

Solution:

The change in flux $= n\left(\phi_2 - \phi_1\right)$ **Total resistance of the circuit** $= 4R + R = 5R$

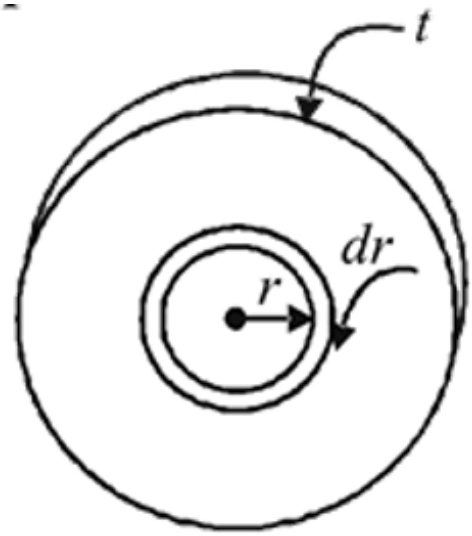
Thus induced current, $i = \frac{\Delta\phi/\Delta t}{R} = \frac{n(\phi_2-\phi_1)}{5RT}$

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Q.4 Answer:

$\frac{\pi t\sigma a^4}{8}\alpha^2$

Solution:



Consider an elemental circle of thickness dr

The induced emf in the circular path of radius r is

$$\varepsilon = \frac{d}{dt}\left(\pi r^2 B\right) = \pi r^2 \alpha$$

The resistance of circular path is

The length of the path being $2\pi r$ and tdr is the cross sectional area of current flow. For the element the power dissipated inside the path is

$$dP = \frac{\varepsilon^2}{R} = \frac{\pi t\sigma}{2}\alpha^2 r^3 dr$$

The total dissipated power P is

$$P = \frac{\pi t\sigma}{2}\alpha^2 \int_0^a r^3 dr = \frac{\pi t\sigma a^4}{8}\alpha^2$$

Q.5 Answer:

$\frac{B_0^2\pi^2 r^4}{R}$

Solution:

$e = -\frac{d\phi}{dt} - -A\frac{dB}{dt} = -\pi r^2 \frac{d}{dt}\left(B_0 e^{-t}\right) = \pi r^2 B_0 e^{-t}$

At $t = 0, e = \pi r^2 B_0, P = \frac{e^2}{R} = \frac{\left(\pi r^2 B_0\right)^2}{R} = \frac{\pi^2 r^4 B_0^2}{R}$

Q.6 Answer:

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I only

Solution:

No solution available

Q.7 Answer:

is counter clockwise

Solution:

As I increases, ϕ increases

$\therefore$ I_i **is such that it opposes the increases in ϕ. Hence, ϕ decreases (By Right Hand Rule). The induced current will be counterclockwise.**

Q.8 Answer:

0.5 weber

Solution:

$$\phi = BA\cos\theta = 2.0 \times 0.5 \times \cos 60^\circ$$
$$= \frac{2.0 \times 0.5}{2} = 0.5 \text{ weber.}$$

Q.9 Answer:

$31\mathrm{mV}$; left to right

Solution:

$\phi = 6\mathrm{t}^2 + 7\mathrm{t} + 1 \Rightarrow \frac{\mathrm{d}\phi}{\mathrm{dt}} = 12\mathrm{t} + 7$

At time, $\mathrm{t} = 2\mathrm{sec}$

$\frac{\mathrm{d}\phi}{\mathrm{dt}} = 24 + 7 = 31\mathrm{mv}$

Direction of current is from left to right according to Flemings right hand rule.

Q.10 Answer:

$\frac{a^2T^3}{3R}$

Solution:

Given that $\phi = \mathrm{at}(\mathrm{T} - \mathrm{t})$

Induced emf, $\mathrm{E} = \frac{\mathrm{d}\phi}{\mathrm{dt}} = \frac{\mathrm{d}}{\mathrm{dt}}[\mathrm{at}(\mathrm{T} - \mathrm{t})]$

$=$ **at** $(0 - 1) + \mathrm{a}(\mathrm{T} - \mathrm{t}) = \mathrm{a}(\mathrm{T} - 2\mathrm{t})$

So, induced emf is also a function of time.

$\therefore$ **Heat generated in time** T **is** $H \int_0^T \frac{E^2}{R} dt = \frac{a^2}{R} \int_0^T (T - 2t)^2 dt = \frac{a^2T^3}{3R}$

Q.11 Answer:

$3.2\pi\mu\mathrm{V}$

Solution:

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Induced emf in the loop is given by $\mathrm{e}=-\mathrm{B} \cdot \frac{\mathrm{dA}}{\mathrm{dt}}$ **where** A **is the area of the loop.**

$e=-B \cdot \frac{d}{d t}\left(\pi r^2\right)=-B \pi 2 r \frac{d r}{d t}$

$\mathrm{r}=2 \mathrm{~cm}=2 \times 10^{-2} \mathrm{~m}$

$\mathrm{dr}=2 \mathrm{~mm}=2 \times 10^{-3} \mathrm{~m}, \mathrm{dt}=1 \mathrm{~s}$

$\mathrm{e}=-0.04 \times 3.14 \times 2 \times 2 \times 10^{-2} \times \frac{2 \times 10^{-3}}{1} \mathrm{~V}$

$=0.32 \pi \times 10^{-5} \mathrm{~V}=3.2 \pi \times 10^{-6} \mathrm{~V}=3.2 \pi \mu \mathrm{V}$

Q.12 Answer:

$0.26 \mathrm{~A}$

Solution:

$e_{\text {ind }}=\frac{d \phi}{d t}=\frac{d}{d t}(B A)=A \frac{d B}{d t}$

$=A \frac{d}{d t}\left(4 t^2+2 t+5\right)=A(8 t+2)=0.06(8 t+2)$

$=2.52$ **at** $t=5$

$i=\frac{2-e_{\text {ind }}}{2}=0.26 A$

Q.13 Answer:

$21.74 \mathrm{~V}$

Solution:

Induced emf. $|\mathrm{e}|=\frac{d \phi}{d t}=A\left(\frac{d B}{d t}\right)$

$=\frac{2 \times 2}{2} \times \frac{d}{d t}(0.042-0.87 t)=2 \times 0.87=1.74 \mathrm{~V}$

The net induced $\mathrm{emf}=20+1.74=21.74 \mathrm{~V}$

Q.14 Answer:

$400 \mathrm{mV}$

Solution:

From Faraday's law, we know that $\mathrm{E}=-\frac{\mathrm{nd} \phi}{\mathrm{dt}}$

$\mathrm{E}=-\mathrm{nA} \frac{\mathrm{dB}}{\mathrm{dt}} \because \cos \theta=1$

$\mathrm{E}=-1000 \times \frac{0.01}{0.01} \times 4 \times 10^{-4}=400 \mathrm{mV}$

Q.15 Answer:

0

Solution:

emf across $\mathrm{AB}=\frac{\pi \mathrm{R}^2}{2} \alpha$

In the upper semicircular loop :

resistance of the across $\mathrm{AB}=\rho \pi \mathrm{R}$

$$=\rho \pi \mathrm{R}$$

current in $\mathrm{AB}=\frac{\text{emf}}{\text{resistance}}=\frac{\pi \mathrm{R}^2 \alpha}{\rho \pi \mathrm{R} 2}=\frac{\alpha \mathrm{R}}{\rho 2}$ **in the clockwise direction.**

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In the lower semicircular loop:

resistance of the across $\mathrm{AB} = \rho\pi\mathrm{R}$

$$= \rho\pi\mathrm{R}$$

current in $\mathrm{AB} = \frac{\text{emf}}{\text{resistance}} = \frac{\pi\mathrm{R}^2\alpha}{\rho\pi\mathrm{R}2} = \frac{\alpha\mathrm{R}}{\rho 2}$ **in the anticlockwise direction.**

Therefore current in $\mathrm{AB} = 0$

Q.16 Answer:

E is constant

Solution:

Let radius of the loop is r **at any time** t **and in further time dt, radius increases by dr. The change in flux** : $\mathrm{d}\phi = (2\pi\mathrm{rdr})\mathrm{B}$

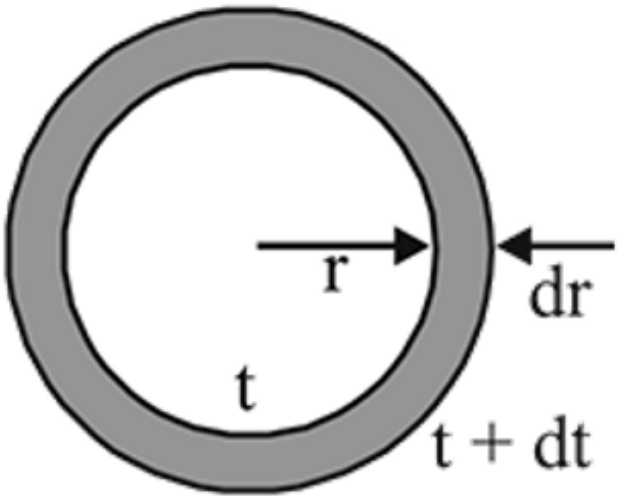
$$\Rightarrow \quad \mathrm{e} = \frac{\mathrm{d}\phi}{\mathrm{dt}} = 2\pi\mathrm{r}\left(\frac{\mathrm{dr}}{\mathrm{dt}}\right)\frac{\mathrm{k}}{\mathrm{r}}$$

$$\Rightarrow \quad \mathrm{e} = 2\pi\mathrm{ck}(\text{constant})$$

$$\left[\because \frac{\mathrm{dr}}{\mathrm{dt}} = \mathrm{c}, \mathrm{B} = \frac{\mathrm{k}}{\mathrm{r}}\right]$$

The change in flux :

$\mathrm{d}\phi = (2\pi\mathrm{rdr})\mathrm{B}$



Q.17 Answer:

zero

Solution:

The magnetic flux through the coil will be zero and so induced emf in it will be zero

Q.18 Answer:

will be equal in both these wheels

Solution:

Since all the wires are connected between rim and axle, they will generate induced emf in parallel, hence it is same for any number of spokes.

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Q.19 Answer:

$\frac{v_0 B_0 d^2}{a}$

Solution:

At $x = 0, B = B_0$, **so** $e_1 = B_0 v_0 d$

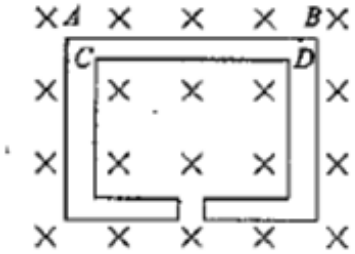
At, $x = d, \quad B = B_0 \left(1 + \frac{d}{a}\right)$, **so** $e_2 = B_0 \left(1 + \frac{d}{a}\right) v_0 d$

Now $e_{net} = e_2 - e_1 = \frac{B_0 v_0 d^2}{a}$

Q.20 Answer:

a to b and d to c

Solution:



Step1:Lenz's Law

Lenz's law, states that an induced electric current flows in a direction such that the current opposes the change that induced it.

Step2:Conclusions

- As the magnetic field magnitude is decreasing, an emf will be induced in the double loop as per Len'z law.

- The direction of this current can be found out using Right hand thumb rule.

- Placing the thumb of our right hand, in the direction of the magnetic field, the direction of other fingers gives us the direction of the flow of current.

- The magnetic field is decreasing in inward direction so according to Len'z law induced current is in clockwise direction due to this induced field is in inward direction.

- As area of outer loop is large so in outer loop current is in clockwise direction and in inner loop it is in anticlockwise direction.

- Here, current flows from $\mathrm{A} \rightarrow \mathrm{B}$ **and** $\mathrm{D} \rightarrow \mathrm{C}$.

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